ALI YOUSEFI

London, United Kingdom | yousefi.ali@icloud.com | +44 7377 253651 | linkedin.com/in/ali-yousefi-6806b

PROFILE

Final-year Computer Science student focused on AI security and applied AI. Lead project is AegisRAG, research on detecting knowledge base poisoning in RAG systems, being prepared for publication with Dr Nikos Komninos. Currently on the Hack The Box AI Red Teamer path alongside CompTIA Security+.

EXPERIENCE

AI Engineering Intern | Adaptive Swarm **2026 – Present**

- Develop and evaluate models on a node-based adaptive AI platform, from data ingestion to signals scored on live markets.
- Give the founders structured feedback as an early engineering user, earning full access to the model and data layer.

Blockchain Security and Market Analyst | Batis Group **2025 – May 2026**

- Screened Ethereum and Tron tokens for rug-pull risk, checking ownership, mint rights, liquidity locks and holder spread.
- Owned company wallet and key management, redesigning storage and recovery to remove single points of failure.

Freelance Web Developer | International B2B Sales Representative, Kaveh Float **2022 – 2025**

- Delivered 2 business websites, including farazjam.com, from brief to launch as sole point of contact.
- Closed 4 international B2B deals across Italy, Germany, China and the Middle East, negotiating terms in a second language.

PROJECTS

Repositories private for security reasons. Walkthrough or live demo available on request.

AegisRAG | Detecting Knowledge Base Poisoning in RAG Systems **2026 – 2027**

Final Year Individual Project | Python, FastAPI, Qdrant, Next.js | Preparing for publication with Dr Nikos Komninos

- Engineered a RAG security testbed over 200 MITRE ATT&CK and CAPEC passages, scored on 44 benchmark questions.
- Showed that synonym-swap poisoning cuts answer correctness by 22% (0.714 to 0.555), mainly by displacing evidence.
- Created a relative-entropy detector reaching 0.95 AUROC that names the swapped word in 22 of 27 flagged passages.
- Halved poisoned text reaching the model (41% to 20% of questions) with a filter that costs no measurable clean accuracy.
- Stress-tested against detector-aware attackers and a second attack, TextFooler, still flagging 85% of its edits.
- Verified all 73 headline claims against 5,000 SHA-256 checksummed run records, backed by 285 passing tests.

AI Lease and Contract Analyser | Retrieval-Augmented Generation **2026**

Python, LLM APIs, embeddings, vector retrieval

- Constructing a RAG pipeline that splits tenancy agreements into clauses and maps each one to the governing statute.
- Flags non-standard, missing and unenforceable terms, and reports accuracy against a hand-labelled test set.
- Scoped in data retention controls and a clear line between analysis and regulated legal advice.

Aris | Voice and Gesture Controlled System Supervisor **2026**

Python, FastAPI, WebSockets, SQLite, OAuth 2.0, MediaPipe

- Created a local assistant with a FastAPI WebSocket backend, a SQLite indexer over 12 git repos and a voice and gesture UI.
- Restricted the tool layer with a repository allowlist and command blocklist to keep every action inside the sandbox.
- Implemented OAuth 2.0 with rotating refresh tokens, tracing a 403 to header filtering and a TLS failure to a missing trust store.

quantdesk | Systematic Trading Research Stack **2026**

Python, DuckDB, Parquet, exchange REST APIs

- Engineered the research harness: exchange data layer, DuckDB and Parquet store, cost model, backtester and walk-forward tests.
- Coded a multi-asset trend strategy with inverse volatility sizing and portfolio volatility targeting on a weekly rebalance.
- Chose rules over machine learning given too few independent market regimes, using walk-forward splits to expose overfitting.

Pharmacy Ordering and Inventory System | Team Software Project **2026**

C#, REST API, MySQL, JWT

- Led requirements, design and sprint planning as Project Manager, and co-wrote the C# front end and JWT-secured REST API.

SECURITY PRACTICE AND CERTIFICATION

In progress: Hack The Box AI Red Teamer path (primary focus), CompTIA Security+ and TryHackMe Red Teaming, due by January 2027.
Completed: TryHackMe Cyber Security 101 path plus web and cloud exploitation rooms.

- **AI red teaming:** practising prompt injection, data poisoning and output manipulation, feeding directly into AegisRAG.
- **Web:** found an exposed .git directory on a Flask host with gobuster and rebuilt the app source from loose objects.
- **Cloud:** abused an AWS Cognito pool issuing credentials to anonymous users to enumerate an over-permissive DynamoDB table.

TECHNICAL SKILLS

Security: LLM red teaming, RAG adversarial testing, web app testing, gobuster, Wireshark, git forensics, cloud IAM, Slither, OAuth 2.0.
Languages: Python, Java, C#, JavaScript, SQL, Bash.
Systems and Cloud: Linux (Kali, Ubuntu), macOS, Windows, TCP/IP, AWS (S3, Cognito, DynamoDB, IAM), Google Cloud, Azure, Docker.
Tools and AI: Git, FastAPI, React, Next.js, Qdrant, REST APIs, MySQL, DuckDB, Parquet, LLM APIs, embeddings, vector search.

EDUCATION

City St George's, University of London | BSc (Hons) Computer Science **2024 – 2027**

Predicted 2:1 or above. Final year: Cloud Computing, Programming and Mathematics for AI, Theory of Computation, 45-credit Individual Project (AegisRAG). Earlier: Networks, Operating Systems, Algorithms.

ADDITIONAL

Languages: English and Turkish (fluent), Persian (native). **Right to work:** eligible for a placement year now, and for the UK Graduate visa (18 months, no sponsorship needed) after graduating in 2027. **Achievements:** Iran U21 National Basketball Champion; UKMT Senior Mathematical Challenge Silver and Senior Kangaroo Certificate of Merit (2022).